9 **(a)** Distinguish between *electromotive force e.m.f.* and *potential difference p.d.* using energy considerations.

E.m.f. is the electrical energy converted from other forms of energy per unit charge delivered by the source. *P.d.*is the electrical energy converted to other forms of energy per unit charge when charges pass from one point to another. [1]

(b) Fig. 9.1 shows a variable resistor and a fixed resistor R_1 connected to a battery with e.m.f E and internal resistance r. The variable resistor takes a range of values between 0 Ω to R_{max}.

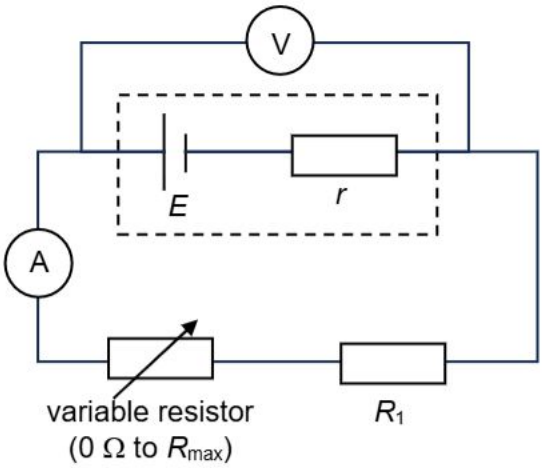


Fig. 9.1

The variation of the voltmeter reading V with current I through the ammeter as the variable resistor is varied over its **full range** of values is shown in Fig. 9.2.

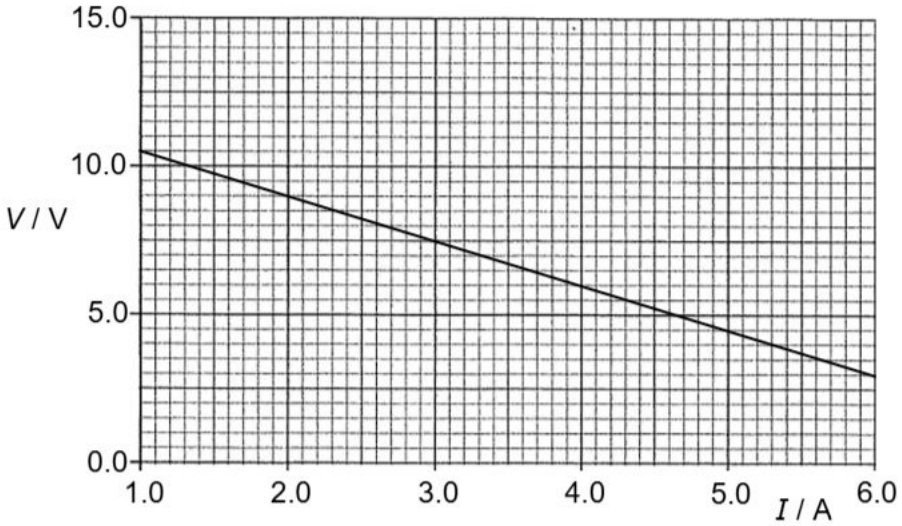


Fig. 9.2

(i) Using the graph, show that that internal resistance r is 1.5 Ω.

Since $E = V + Ir$ → $V = -rI + E$ (in the form of y = mx + c)

gradient of V–I graph gives **negative** value of internal resistance r

[1 identify which part of graph is related to r]

gradient = (10.5 – 3.0)/(1.0–6.0) [1 for correct coordinates on line, far apart]

gradient = –1.5 Ω

$\therefore r$ = 1.5 Ω [1]

[3]

(ii) Calculate the e.m.f. E of the battery.

When V = 10.5 V, I = 1.0 A

Using $E = V + Ir$

= 10.5 + 1.0(1.5) = 12.0 V

Note: E is **not** 10.5 V, as current at that point is not 0 A. Hence not y-intercept

e.m.f. E = 12.0 V [1]

(iii) Using Fig. 9.2, calculate the maximum resistance of the variable resistor R_{max}. Explain your working.

When variable resistor is set to 0 Ω, the resistance in the circuit is the lowest, hence current will be the highest. V = 3.0 V, I = 6.0 A

p.d. across R_1 is equal to V.

Using $V = IR_1$,

$R_1 = V / I$ = 3.0/6.0 = 0.50 Ω [1]

When variable resistor is set to max, the current will be the lowest.
V = 10.5 V, I = 1.0 A ,

Using $V = IR$, where R is the total resistance of R_{max} and R_1

$R_{max} + R_1 = V / I$

R_{max} + 0.50 = 10.5 / 1.0

R_{max} = 10.0 Ω [1]

(1 mark for explanation of choices of points)

resistance R_{max} = ... Ω [3]

(c) The variable resistor is now set to 6.0 Ω, and a thermistor is connected in parallel with the variable resistor and R_1. The ammeter is also now connected in series with the thermistor as shown in Fig. 9.3.

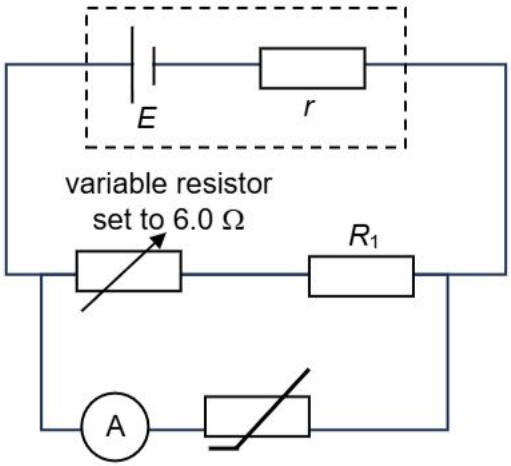


Fig. 9.3

(i) Explain how the potential difference across the thermistor changes when the temperature rises.

When the temperature increases, the resistance of the thermistor decreases. [1]

Hence the effective resistance of the external resistors and the thermistor, which are in parallel, decreases. By potential divider principle, the potential difference across both the external resistor and thermistor decreases. [1]

[2]

(ii) Hence, or otherwise, determine how the current across the thermistor changes when the temperature rises.

Since the effective resistance of the two external resistors and the thermistor decreases, the total resistance of the circuit decreases. Which means that the total current I_{total} exiting the cell increases.[1] Since the potential difference across the two external resistors decreases, by $V=IR$ the current I_R across the two external resistors decreases. [1]

Since $I_{total} = I_R + I_{Thermistor}$, the current through the thermistor increases. [1]

[3]

(d) Fig. 9.4 shows an alternating input voltage V_{IN} connected to a capacitor C and a resistor R.

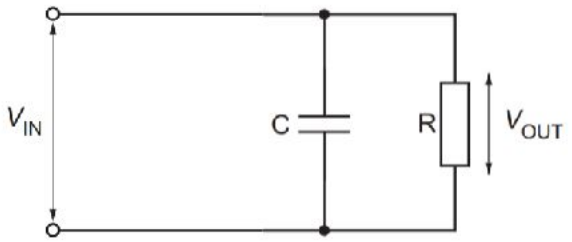


Fig. 9.4

The input voltage V_{IN} is a square wave. Fig 9.5 shows the variation of V_{IN} with time t.

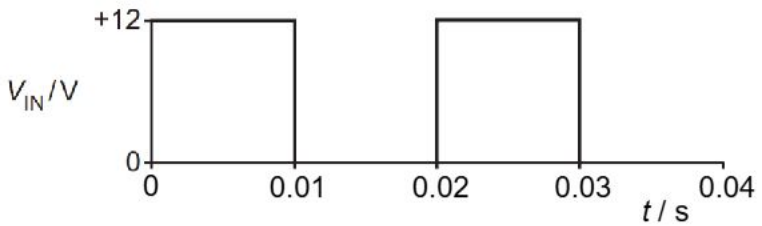


Fig. 9.5

Fig. 9.6 shows the variation of V_{OUT} with time t.

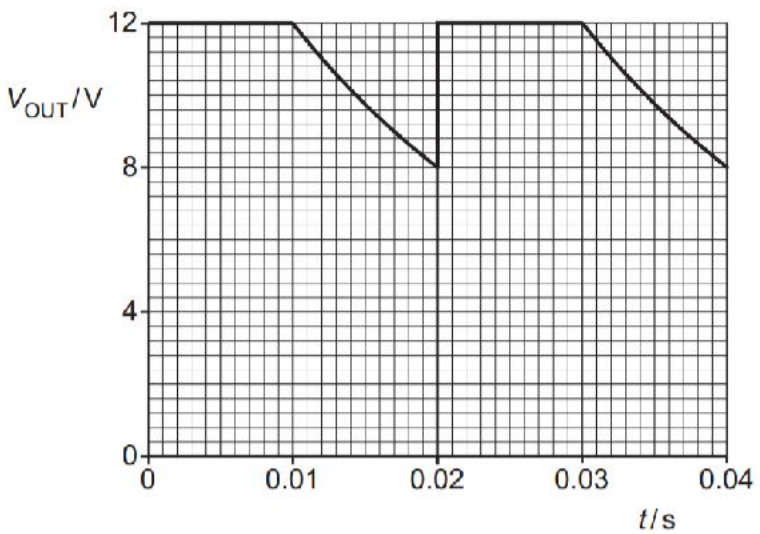


Fig. 9.6

The maximum energy stored in the capacitor is 0.041 J.

(i) Show that the capacitance of C is 570 μF.

Energy stored in a capacitor $E = \frac{1}{2}CV^2$

$C = 2E/V^2 = \frac{2 \times 0.041}{12^2}$ [1]

$C = 569.44 \times 10^{-6}$ [1]

$= 570\ \mu F$

[2]

(ii) Determine the resistance of R.

It takes 0.010 s for p.d. to drop from 12.0 V to 8.0 V

Using $V = V_0 e^{-\frac{t}{\tau}}$, where $\tau = RC$

$8.0 = (12.0)\, e^{-\frac{0.010}{RC}}$ [1]

$$\ln(8.0/12.0) = -\frac{0.010}{R \times 570 \times 10^{-6}}$$

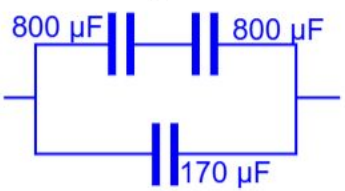
$R = 43.27\,\Omega = 43\,\Omega$ [1]

(note1: while not a marking pt, time is read to half a square precision, 0.010s)

(note2: recommended to use the proven value of C)

resistance = ... Ω [2]

(iii) Given two capacitors rated 800 μF and a third capacitor of 170 μF, suggest how capacitor *C* of 570 μF can be constructed from these three capacitors. Include a labelled diagram and show your workings clearly.



1m for diagram

Capacitance of two 800 μF in series = $\left(\frac{1}{800} + \frac{1}{800}\right)^{-1} = 400$ μF [1]

Capacitance of 400 μF and 170 μF in parallel = 400 + 170 = 570 μF [1]

[3]

[Total: 20]